EE561 Advanced Static Power Conversion · Multilevel Converters module

Potential Final Questions: Building the Levels

Proposed by Sefer İdacı for the final examination

Two questions for a closed-book exam, each designed for about 12 minutes and 25 points, on the fundamentals of the topologies talk: levels and device voltage with the NPC leg, and the states of a flying-capacitor leg. No derivations are asked; every part needs a short table, one number or one sentence. Solutions and a marking guide follow. All numbers differ from the lecture examples.

Question 1 · Levels, device voltage and the NPC leg

FUNDAMENTALS **25 pts**

A converter has a DC link of V_{dc} = **1200 V**. In an N-level leg each switch blocks $V_{dc}/(N-1)$. A switch may block at most **2/3 of its voltage rating**; the available classes are 650 V, 1200 V, 1700 V and 3300 V.

a. For N = 2, 3 and 5, give the voltage blocked by each switch and the smallest suitable device class. [9]

b. For a 3-level NPC leg (switches S1–S4 from top to bottom, clamp diodes D5 and D6), list the switches that are on and the output voltage v_{aO} in states P, O and N. [6]

c. The leg is in state O. Which two devices carry the current when it flows out of the leg ($i_a > 0$), and which two when it flows into the leg ($i_a < 0$)? [6]

d. Give one reason why the 3-level leg has lower switching loss than the 2-level leg at the same switching frequency, and one disadvantage of the 3-level leg. [4]

Question 2 · States of a flying-capacitor leg

FUNDAMENTALS **25 pts**

The 3-level flying-capacitor leg in the figure has V_{dc} = 1000 V and its flying capacitor is held at V_{fc} = 500 V. Switches S1/S1′ and S2/S2′ are complementary pairs. The current flows out of the leg ($i_a > 0$).

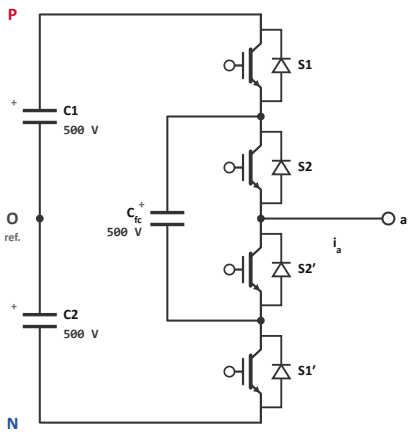
a. For the four states S1 S2 = 11, 10, 01, 00, give the output voltage v_{aO} and say whether the flying capacitor is charged, discharged or bypassed. [8]

b. Which two states are redundant, and what is the choice between them used for? [5]

c. V_{fc} is measured as 470 V, which is too low, and the next output level is 0 V. Which state do you apply? [4]

d. The capacitor ripple is $\Delta V = \hat{I} / (2 f_{sw} C_{fc})$. Find C_{fc} for ΔV = 10 V at $\hat{I}$ = 60 A and f_{sw} = 20 kHz. [4]

e. Why must C_{fc} be charged before the leg starts switching? [4]



3-level flying-capacitor leg (O: reference midpoint only).

Solutions and marking guide

Solution 1 · Levels, device voltage and the NPC leg — 25 pts

(a) Needed rating = 1.5 × blocked voltage; take the next class. [9: 1.5 per cell]

Levels N	Blocked voltage	Needed rating	Class
2	1200 V	1800 V	**3300 V**
3	600 V	900 V	**1200 V**
5	300 V	450 V	**650 V**

(b) [6: 1 per cell]

State	Switches on	v_{aO}
P	S1, S2	+600 V
O	S2, S3	0 V
N	S3, S4	−600 V

(c) $i_a > 0$: **D5 and S2** (from the midpoint through the upper clamp diode). $i_a < 0$: **S3 and D6** (through the lower clamp diode into the midpoint). [6: 3 + 3]

(d) Lower switching loss: each commutation switches only $V_{dc}/2$, and the lower-voltage device is faster (less energy per switching event). Disadvantage, any one: twice the switches and gate drivers; two devices in series (more conduction loss); the midpoint voltage must be balanced; more states, harder control. [4: 2 + 2]

Solution 2 · States of a flying-capacitor leg — 25 pts

(a) [8: 1 per cell]

S1 S2	v_{aO}	Flying capacitor ($i_a > 0$)
11	+500 V	bypassed
10	0 V (= 500 − V_{fc})	charged
01	0 V (= −500 + V_{fc})	discharged
00	−500 V	bypassed

(b) 10 and 01: both give 0 V. One charges the flying capacitor and the other discharges it, so choosing between them keeps V_{fc} at its target without changing the output. [5: 2 + 3]

(c) The capacitor must be charged, and with $i_a > 0$ that is **state 10**. [4]

(d) $C_{fc} = \hat{I}/(2\, f_{sw}\, \Delta V) = 60/(2 \cdot 20\,000 \cdot 10) =$ **150 µF**. [4]

(e) With an empty capacitor the switches no longer share the voltage: one switch would have to block the full V_{dc} = 1000 V instead of $V_{dc}/2$, and would fail. The output levels would also be wrong. [4]